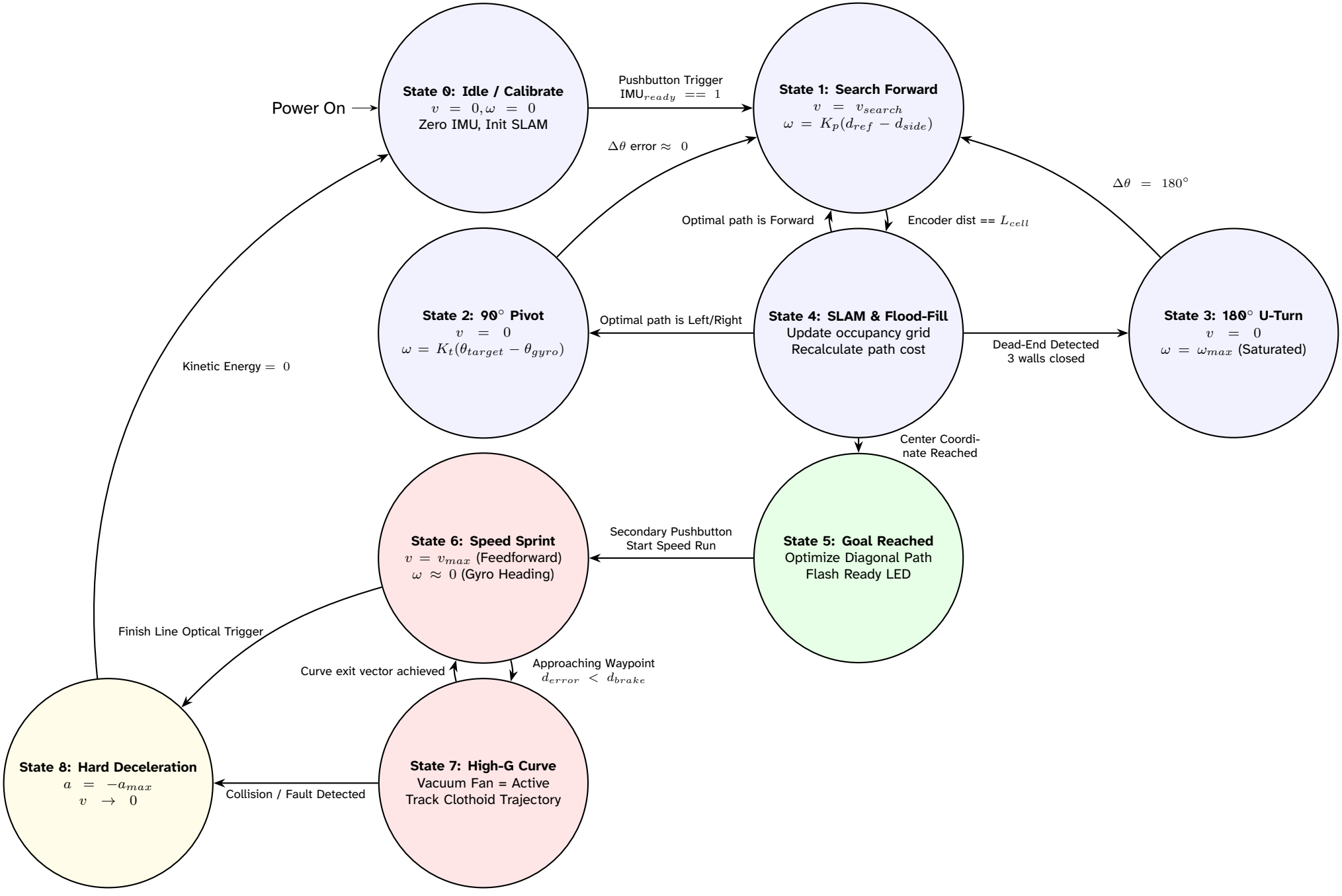

Power On
State 0: Idle / Calibrate
$v = 0, \omega = 0$
Zero IMU, Init SLAM
Pushbutton Trigger
$\text{IMU}_{ready} == 1$
State 1: Search Forward
$v = v_{search}$
$\omega = K_p(d_{ref} - d_{side})$
$\Delta\theta$ error ≈ 0
$\Delta\theta = 180^\circ$
Optimal path is Forward
Encoder dist == L_{cell}
State 2: 90° Pivot
$v = 0$
$\omega = K_t(\theta_{target} - \theta_{gyro})$
Optimal path is Left/Right
State 4: SLAM & Flood-Fill
Update occupancy grid
Recalculate path cost
Dead-End Detected
3 walls closed
State 3: 180° U-Turn
$v = 0$
$\omega = \omega_{max}$ (Saturated)
Kinetic Energy $= 0$
Center Coordinate Reached
State 6: Speed Sprint
$v = v_{max}$ (Feedforward)
$\omega \approx 0$ (Gyro Heading)
Secondary Pushbutton
Start Speed Run
State 5: Goal Reached
Optimize Diagonal Path
Flash Ready LED
Finish Line Optical Trigger
Curve exit vector achieved
Approaching Waypoint
$d_{error} < d_{brake}$
State 8: Hard Deceleration
$a = -a_{max}$
$v \rightarrow 0$
Collision / Fault Detected
State 7: High-G Curve
Vacuum Fan = Active
Track Clothoid Trajectory